

Latika Liladhar Dekate

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EDUCATION

Arizona State University

M.S., Data Science, Analytics & Engineering [Electrical Engineering] (GPA: 3.89/4.00)

Tempe, AZ

Graduating Dec 2026

S.I.E.S. Graduate School of Technology

B.E., Electronics & Telecommunication Engineering (GPA: 3.71/4.00)

Mumbai, IN

Aug 2020 – Jun 2024

TECHNICAL SKILLS

AI & LLMs: LangChain, GraphRAG, Hugging Face Transformers (FinBERT, DistilBERT, Llama 3.1), RAG Pipelines, Agentic Workflows, Prompt Engineering

ML: PyTorch, TensorFlow, Scikit-learn, Deep Learning, Feature Engineering, Anomaly Detection, Model Evaluation

Programming & Data: Python (NumPy, pandas, multiprocessing), SQL/PostgreSQL, PySpark, Bash, R

Data Engineering: ETL Pipelines, Apache Kafka, Neo4j, ChromaDB, Docker, GCP

Visualization & Tools: Streamlit, Flask, D3.js, Tableau, Power BI, Git, Linux, Kubernetes

PROJECTS

Stealthy Data Poisoning in RAG Systems | *Python, LangChain, ChromaDB, Llama 3.1*

Mar 2026 – Apr 2026

- Designed a controlled security evaluation framework over a 5,000-document RAG pipeline, testing 5 stealthy poisoning variants against 1,000 benchmark queries to quantify retrieval degradation under adversarial conditions.
- Measured near-random anomaly separability (AUC 0.527) across attack types, identifying Top-K capping as the most effective mitigation strategy and producing a reproducible evaluation protocol for future RAG security benchmarks.

AstraNode : GraphRAG Research Intelligence | *Neo4j, GraphRAG, LangChain, Python*

Oct 2025 – Nov 2025

- Architected a graph-based retrieval system over 600+ NASA research papers using Neo4j and LangChain, replacing keyword search with semantic graph and surfacing cross-mission connections previously inaccessible through traditional indexing.
- Reduced research discovery time by 40% across complex multi-domain queries; delivered full pipeline documentation for reproducibility and won the Arizona Regional NASA Space Apps Challenge.

Advanced Text Analyzer | *Flask, REST API, D3.js, Full-Stack*

Aug 2025 – Sep 2025

- Built a full-stack NLP application unifying 10,000+ records from CSV and JSON sources, deploying domain-specific transformer models (FinBERT for finance, DistilBERT for reviews) with 95%+ accuracy across three classification domains.
- Integrated Named Entity Recognition via SpaCy and a Flask REST API with a D3.js dashboard, providing stakeholders with traceable sentiment trends and severity metrics through a single interface.

EXPERIENCE

Arizona State University, DaRL Lab, SCAI

Tempe, AZ

Research Intern, Sim2Real Traffic Signal Control Benchmark

May 2026 – Present

- Synthesizing recent RL literature on action-delay and sim-to-real action gaps, comparing prediction-based and action-history-based methods to guide benchmark design for traffic signal control.
- Inspecting the Sim2RealTSCBenchMark repository to understand agent, trainer, and environment action flow, supporting planned implementation of delayed signal-action experiments.

Google Developer Student Clubs, S.I.E.S. Graduate School of Technology

Mumbai, IN

ML & Data Science Program Coordinator

Jul 2022 – Apr 2024

- Designed and delivered a 10-week ML curriculum covering PyTorch, TensorFlow, and production deployment practices for 150+ students, increasing project completion rates by 15% through structured mentoring and iterative feedback.
- Directed cross-functional student teams through the end-to-end ML lifecycle, from data preprocessing to model deployment, building a cohort capable of shipping production-grade projects independently.

S.I.E.S. Graduate School of Technology

Mumbai, IN

Machine Learning & Image Processing Intern

Jun 2022 – Jul 2022

- Diagnosed and debugged failure cases in an object detection pipeline by tuning normalization, cropping, and augmentation parameters in Python, improving held-out performance by 10% and reducing label noise in production review.
- Addressed class imbalance on a credit card fraud dataset by benchmarking sampling strategies and tree-based classifiers, selecting a final model achieving an F1 score of 0.92 aligned to product risk thresholds.

CERTIFICATIONS & RECOGNITION

Patent: Multilayer Microstrip Aerial Transmitter Receptor for Wi-Fi Applications

Apr 2024

NASA Space Apps Challenge : Regional Winner & Global Nominee

Oct 2025

Google Cloud Career Practitioners Certificate : Cloud Engineering & Data Science / ML Tracks

Jan 2023